

AMAN KUMAR SINGH

+91-6203553465 akamansingh07@gmail.com LinkedIn Github Portfolio

Summary

Software Engineer with strong expertise in Core Java, Spring Boot, RESTful APIs, SQL, and Backend Development. Experienced in developing scalable backend applications using Spring Boot and relational databases. Proficient in object-oriented programming, authentication, API development, and software engineering best practices with a passion for building secure and maintainable enterprise applications.

Experience

Healthletic Lifestyle

Remote

Backend Engineering Intern

Jun 2026 – Present

- Developed and maintained scalable RESTful APIs using Node.js, Express.js, and MongoDB, implementing core business logic, database operations, JWT-based authentication and authorization, and request validation to deliver secure and reliable backend services.
- Designed, tested, and documented REST APIs using Swagger and Postman; integrated third-party services and ensured seamless communication between frontend and backend systems across development and production environments.
- Collaborated with cross-functional teams in an Agile environment to troubleshoot backend issues, implement new features, support deployments, participate in code reviews, and manage version control using Git.
- Optimized API response times and database queries in MongoDB, improving overall application performance and reliability.

Education

Master of Computer Application

PG: 2026

Sharda University, Uttar Pradesh

CGPA: 9.27 / 10

Bachelor of Computer Application

UG: 2024

JIS College of Engineering, West Bengal

CGPA: 8.14 / 10

Projects

EduNex-LMS | Java 21, Spring Boot, Spring Data JPA, MySQL, Docker

July 2026

- Engineered a production-style Learning Management System (LMS) backend using Java 21, Spring Boot 4.1, Spring Data JPA, Hibernate, and MySQL with a feature-based modular layered architecture.
- Designed a stateless JWT authentication system with database-backed refresh token rotation, reuse detection, and Spring Security-based role and resource-level authorization.
- Developed scalable REST APIs with CRUD operations, pagination, sorting, and dynamic filtering using JPA Specifications, while optimizing database performance through @EntityGraph to eliminate N+1 queries.
- Built production-ready APIs using DTOs with MapStruct, Bean Validation, centralized exception handling, Swagger/OpenAPI, and containerized the application with Docker and Docker Compose.

Secure Notes | Spring Boot, Spring Security, MySQL, Docker

Jun 2024

- Developed a secure Notes Management REST API using Java, Spring Boot, Spring Data JPA, Hibernate, and MySQL, following a layered architecture with clean separation of concerns.
- Implemented JWT-based authentication and authorization with Spring Security, enabling secure user registration, login, and protected CRUD operations.
- Built REST APIs with DTOs, Bean Validation, centralized exception handling, pagination, and Swagger/OpenAPI for robust and well-documented API development.
- Containerized the application using Docker and managed data persistence through Spring Data JPA/Hibernate, ensuring maintainable and scalable backend architecture.

Heart-AI: Dual-Engine Cardiovascular Diagnostic | Python, Flask, TensorFlow, XGBoost

Mar 2026

- Engineered a dual-engine cardiovascular diagnostic platform utilizing Flask, XGBoost, and CNN-based deep learning to concurrently analyze tabular clinical vitals and ECG image scans.
- Trained and deployed machine learning models on clinical datasets, achieving 92% prediction accuracy for real-time heart disease risk assessment and classification.
- Designed a full-stack analytics dashboard integrating SQLite, Chart.js, and FPDF to provide interactive health visualizations, patient record management, and automated PDF report generation.
- Developed secure RESTful APIs and integrated machine learning inference with the Flask backend, enabling real-time predictions and seamless communication between the web interface and AI models.

Technical Skills

- Programming Languages:** Java, JavaScript, Python, C, C++, SQL
- Frameworks & Libraries:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Node.js, Express.js
- Backend Development:** RESTful APIs, JWT Authentication, API Documentation (Swagger/OpenAPI), MapStruct, Bean Validation
- Databases:** MySQL, PostgreSQL, MongoDB
- Tools & Platforms:** Git, GitHub, Docker, Maven, Postman, AWS, IntelliJ IDEA, VS Code
- Core Concepts:** Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), Design Patterns, MVC Architecture, Exception Handling
- Development Practices:** Agile (Scrum), Git Workflow, Code Reviews